

0 3 . 1 In taxonomy, an organism is identified by referring to the species name and the genus name.

What term is used to describe this method of naming organisms?

[1 mark]

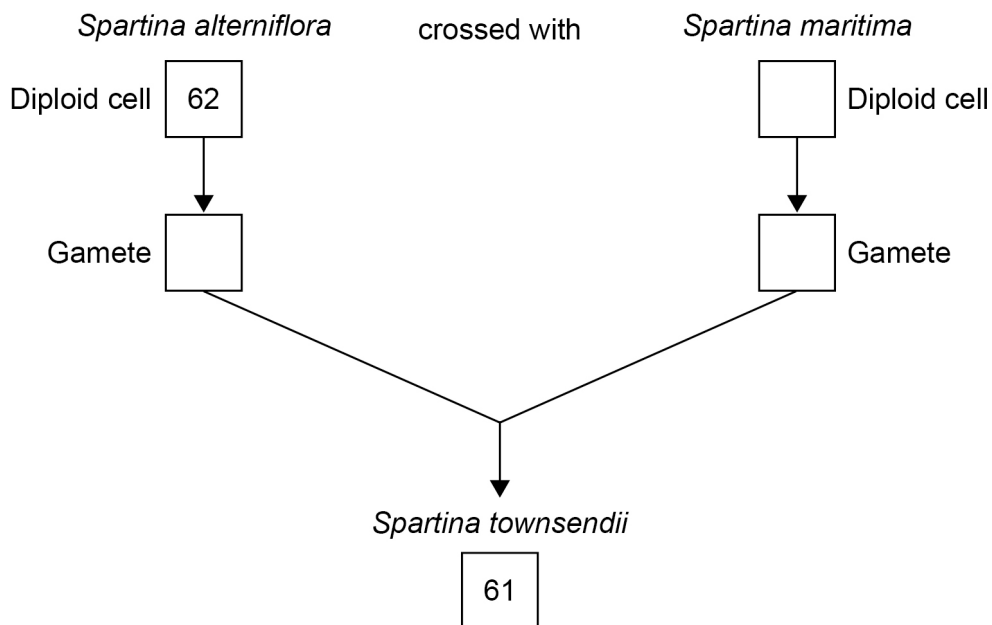
0 3 . 2 Define the term mutagenic agent.

[1 mark]

0 3 . 3 **Figure 4** shows how the species *Spartina townsendii* is produced.

The number of chromosomes in cells is shown in some of the boxes.

Figure 4



Complete **Figure 4** by giving the correct number of chromosomes in each of the boxes.

[1 mark]

A mutation in the number of chromosomes in a *S. townsendii* cell produced a new species, *Spartina anglica*.

Figure 5 shows the number of chromosomes in leaf cells of these species.

Figure 5

S. townsendii

61

S. anglica

122

0 3 . 4

Name the type of mutation that changed the number of chromosomes in *S. townsendii* to produce *S. anglica*. Explain your answer.

[3 marks]

Name of mutation _____

Explanation _____

0 3 . 5

Genetic variation within a species is increased during meiosis by crossing over and the independent segregation of homologous chromosomes.

Apart from mutation, explain **one** other way genetic variation within a species is increased.

[2 marks]

